ITSimplera Institute

WEEK 03: Multi-Protocol Routing: OSPF, RIP, and Static Route Redistribution in GNS3

Domain: Network administration

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Multi-Protocol Routing

Routing Protocols:

A routing protocol specifies how routers communicate with each other to distribute information that enables them to select paths between nodes on a computer network

Routing protocols can be broken down into a few main categories depending on their function and scale:

- **OSPF (Open Shortest Path First):** A widely used, advanced protocol that calculates the shortest and fastest path to a destination using mathematical algorithms.
- **RIP (Routing Information Protocol):** A simpler, older protocol that determines the best path based simply on the lowest "hop count" (the number of routers a packet must pass through).
- **Static routing:** A process where a network administrator manually configures and defines the exact paths data packets must take to reach a specific destination network.

GNS3 Multi-Protocol Routing Topology:

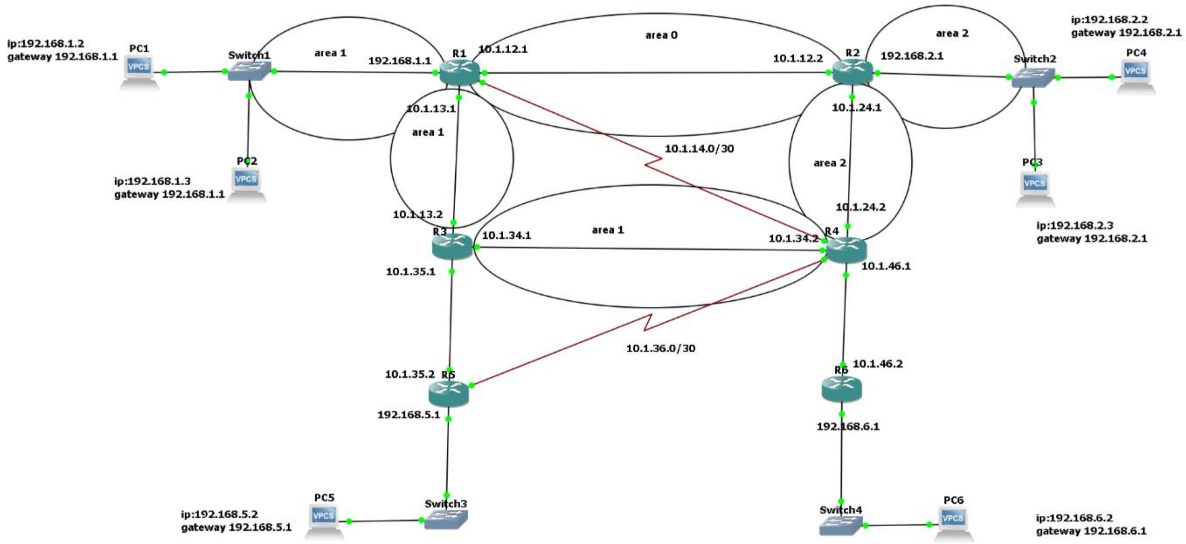

figure 1.0(complete network topology)

Here are what protocols are used in this network:

Router	Protocol(s)	Special Role
R1	OSPF (Area 0/1)	ABR + static backup link to R4
R2	OSPF (Area 0/2)	ABR
R3	OSPF (Area 1) + RIPv2	RIP↔OSPF redistribution point
R4	OSPF (Area 1/2) + Static	ABR + Static↔OSPF redistribution point + 2 floating static backups
R5	RIPv2	Floating static backup link to R4
R6	Static only	No dynamic protocol; default route to R5

IP Addressing Tables:

Router-to-Router Links (Serial/WAN):

Link	Router A	Interface IP (A)	Router B	Interface IP (B)	Network	Type
1	R1	10.1.12.1	R2	10.1.12.2	10.1.12.0/30	Primary
2	R1	10.1.13.1	R3	10.1.13.2	10.1.13.0/30	Primary
3	R2	10.1.24.1	R4	10.1.24.2	10.1.24.0/30	Primary
4	R3	10.1.34.1	R4	10.1.34.2	10.1.34.0/30	Primary
5	R3	10.1.35.1	R5	10.1.35.2	10.1.35.0/30	Primary
6	R4	10.1.46.1	R6	10.1.46.2	10.1.46.0/30	Primary

Backup Links: Shown with the serial connection (in Red):

Link	Router A	Router B	Network	Purpose
Backup 1	R1	R4	10.1.14.0/30	Backup link between Area 1 (R1) and Area 2 (R4)
Backup 2	R5	R4	10.1.36.0/30	Backup link between R5 (Area 1) and R4 (Area 2)

Local Area Networks (LANs):

LAN	Router	Router Interface IP	Switch	Attached PCs	PC IP	PC Gateway
LAN 1	R1	192.168.1.1	Switch1	PC1, PC2	192.168.1.2 (PC1), 192.168.1.3 (PC2)	192.168.1.1
LAN 2	R2	192.168.2.1	Switch2	PC3, PC4	192.168.2.3 (PC3), 192.168.2.2 (PC4)	192.168.2.1
LAN 3	R5	192.168.5.1	Switch3	PC5	192.168.5.2	192.168.5.1
LAN 4	R6	192.168.6.1	Switch4	PC6	192.168.6.2	192.168.6.1

Router Configurations:

R1(router 1):

```
interface FastEthernet0/1 ##link config
 ip address 10.1.13.1 255.255.255.252

interface Serial0/1 ##backup link
 ip address 10.1.14.1 255.255.255.252
```

```
interface FastEthernet1/0
 ip address 192.168.1.1 255.255.255.0

router ospf 1 #main links
 router-id 1.1.1.1
 network 10.1.12.0 0.0.0.3 area 0
 network 10.1.13.0 0.0.0.3 area 1
 network 192.168.1.0 0.0.0.255 area 1

ip route 192.168.2.0 255.255.255.0 10.1.14.2 150 ## backup links
ip route 192.168.6.0 255.255.255.0 10.1.14.2 150
```

R2(Router 2):

```
interface FastEthernet0/0
 ip address 10.1.12.2 255.255.255.252

interface FastEthernet0/1
 ip address 10.1.24.1 255.255.255.252

interface FastEthernet1/0
 ip address 192.168.2.1 255.255.255.0

router ospf 1
 router-id 2.2.2.2
 network 10.1.12.0 0.0.0.3 area 0
 network 10.1.24.0 0.0.0.3 area 2
 network 192.168.2.0 0.0.0.255 area 2
```

Router 3:

```
interface FastEthernet0/0
 ip address 10.1.35.1 255.255.255.252

interface FastEthernet0/1
 ip address 10.1.13.2 255.255.255.252

interface FastEthernet1/0
 ip address 10.1.34.1 255.255.255.252

router ospf 1
 router-id 3.3.3.3
 network 10.1.13.0 0.0.0.3 area 1
 network 10.1.35.0 0.0.0.3 area 1

router rip
 version 2
 redistribute ospf 1 metric 3
 network 10.0.0.0
 no auto-summary
```

Router 4:

```
ip address 10.1.46.1 255.255.255.252

interface Serial0/0
 ip address 10.1.36.2 255.255.255.252
```

```
interface FastEthernet0/1
 ip address 10.1.24.2 255.255.255.252

interface Serial0/1
 ip address 10.1.14.2 255.255.255.252

interface FastEthernet1/0
 ip address 10.1.34.2 255.255.255.252

router ospf 1
 router-id 4.4.4.4
network 10.1.24.0 0.0.0.3 area 2
 network 10.1.34.0 0.0.0.3 area 1

ip route 192.168.5.0 255.255.255.0 10.1.36.1 150
ip route 192.168.6.0 255.255.255.0 10.1.46.2
```

Router 5:

```
interface FastEthernet0/0
 no ip address
 shutdown

interface Serial0/0
 ip address 10.1.36.1 255.255.255.252

interface FastEthernet0/1
 ip address 10.1.35.2 255.255.255.252

interface FastEthernet1/0
 ip address 192.168.5.1 255.255.255.0

router rip
 version 2
 network 10.0.0.0
 network 192.168.5.0
 no auto-summary

ip route 192.168.2.0 255.255.255.0 10.1.36.2 130
ip route 192.168.6.0 255.255.255.0 10.1.36.2 130
```

Router 6:

```
interface FastEthernet0/0
 ip address 10.1.46.2 255.255.255.252

interface FastEthernet1/0
 ip address 192.168.6.1 255.255.255.0


ip route 0.0.0.0 0.0.0.0 10.1.46.1
```

IP Routing Tables:

These screen show complete routing tables and Redistribution routes from OSPF to RIPv2 and Static Routing

Router 1:

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E1 192.168.5.0/24 [110/40] via 10.1.12.2, 00:00:16, FastEthernet0/0
     10.0.0.0/30 is subnetted, 7 subnets
C       10.1.14.0 is directly connected, Serial0/1
C       10.1.13.0 is directly connected, FastEthernet0/1
C       10.1.12.0 is directly connected, FastEthernet0/0
O IA    10.1.24.0 [110/20] via 10.1.12.2, 00:00:16, FastEthernet0/0
O       10.1.35.0 [110/20] via 10.1.13.2, 00:00:16, FastEthernet0/1
O E1    10.1.34.0 [110/30] via 10.1.13.2, 00:00:16, FastEthernet0/1
O E1    10.1.36.0 [110/40] via 10.1.13.2, 00:00:18, FastEthernet0/1
O E1 192.168.6.0/24 [110/40] via 10.1.12.2, 00:00:18, FastEthernet0/0
C    192.168.1.0/24 is directly connected, FastEthernet1/0
O IA 192.168.2.0/24 [110/11] via 10.1.12.2, 00:00:18, FastEthernet0/0
```

This screenshot includes the routing codes

Router 2:

```
R2#show ip rou
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E1 192.168.5.0/24 [110/30] via 10.1.24.2, 00:02:01, FastEthernet0/1
     10.0.0.0/30 is subnetted, 6 subnets
O IA    10.1.13.0 [110/20] via 10.1.12.1, 00:02:01, FastEthernet0/0
C       10.1.12.0 is directly connected, FastEthernet0/0
C       10.1.24.0 is directly connected, FastEthernet0/1
O IA    10.1.35.0 [110/30] via 10.1.12.1, 00:01:57, FastEthernet0/0
O E1    10.1.34.0 [110/40] via 10.1.12.1, 00:01:51, FastEthernet0/0
O E1    10.1.36.0 [110/50] via 10.1.12.1, 00:01:51, FastEthernet0/0
O E1 192.168.6.0/24 [110/30] via 10.1.24.2, 00:02:03, FastEthernet0/1
O IA 192.168.1.0/24 [110/11] via 10.1.12.1, 00:02:03, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet1/0
R2#
```

Router 3:

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.5.0/24 [150/0] via 10.1.36.1
     10.0.0.0/30 is subnetted, 8 subnets
C       10.1.14.0 is directly connected, Serial0/1
O IA    10.1.13.0 [110/30] via 10.1.24.1, 00:03:47, FastEthernet0/1
O IA    10.1.12.0 [110/20] via 10.1.24.1, 00:03:47, FastEthernet0/1
C       10.1.24.0 is directly connected, FastEthernet0/1
C       10.1.46.0 is directly connected, FastEthernet0/0
O IA    10.1.35.0 [110/40] via 10.1.24.1, 00:03:43, FastEthernet0/1
C       10.1.34.0 is directly connected, FastEthernet1/0
C       10.1.36.0 is directly connected, Serial0/0
S    192.168.6.0/24 [1/0] via 10.1.46.2
O IA 192.168.1.0/24 [110/21] via 10.1.24.1, 00:03:49, FastEthernet0/1
O    192.168.2.0/24 [110/11] via 10.1.24.1, 00:03:49, FastEthernet0/1
R4#
```

Router 5:

```
R5#show ip rou
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.5.0/24 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 6 subnets
R       10.1.13.0 [120/1] via 10.1.35.1, 00:00:06, FastEthernet0/1
R       10.1.12.0 [120/3] via 10.1.35.1, 00:00:06, FastEthernet0/1
R       10.1.24.0 [120/3] via 10.1.35.1, 00:00:06, FastEthernet0/1
C       10.1.35.0 is directly connected, FastEthernet0/1
R       10.1.34.0 [120/1] via 10.1.35.1, 00:00:06, FastEthernet0/1
C       10.1.36.0 is directly connected, Serial0/0
R    192.168.6.0/24 [120/3] via 10.1.35.1, 00:00:08, FastEthernet0/1
R    192.168.1.0/24 [120/3] via 10.1.35.1, 00:00:08, FastEthernet0/1
R    192.168.2.0/24 [120/3] via 10.1.35.1, 00:00:08, FastEthernet0/1
```

Router 6:

```
R6#
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.46.1 to network 0.0.0.0

     10.0.0.0/30 is subnetted, 1 subnets
C       10.1.46.0 is directly connected, FastEthernet0/0
C    192.168.6.0/24 is directly connected, FastEthernet1/0
S*   0.0.0.0/0 [1/0] via 10.1.46.1
R6#[~
```

OSPF neighbour verification

Router 1:

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:31    10.1.12.2       FastEthernet0/0
3.3.3.3           1   FULL/DR         00:00:31    10.1.13.2       FastEthernet0/1
R1#
```

Router 2:

```
R2#SHOW IP OSpf NEIGhbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:33    10.1.12.1       FastEthernet0/0
4.4.4.4           1   FULL/DR         00:00:34    10.1.24.2       FastEthernet0/1
R2#
```

Router 3:

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:34    10.1.13.1       FastEthernet0/1
R3#
```

Router 4:

```
R4#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:35    10.1.24.1       FastEthernet0/1
R4#
```

Route Tracing from Virtual PCs:

Tracing Route from LAN1 PC1 to LAN4 shows the number of hops, in this case from PC 1 to PC 6 there were 5 hops each hop has its time written in milliseconds, while tracing routes from PC-1 to PC 3 shows that there were only 2 hops, because LAN 1 is close to LAN 3

```
PC1> trace 192.168.6.2
trace to 192.168.6.2, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   14.684 ms  14.824 ms  16.057 ms
 2   10.1.12.2   45.302 ms  46.187 ms  46.153 ms
 3   10.1.24.2   75.985 ms  75.712 ms  77.302 ms
 4     *10.1.46.2   107.351 ms  105.488 ms
 5     **192.168.6.2   124.528 ms (ICMP type:3, code:3, Destination port unreach
able)

PC1> trace 192.168.2.1
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   15.480 ms  15.606 ms  15.066 ms
 2   *10.1.12.2   46.919 ms (ICMP type:3, code:3, Destination port unreachable)
```

Tracing Route from LAN3 PC5 to LAN2 PC 3 shows the number of hops, in this case there were 5 hops, with time took for each hop written in milliseconds. While tracing route from LAN 3 PC5 to LAN 4 PC6 there were 8 hops

```
trace 192.168.2.3
trace to 192.168.2.3, 8 hops max, press Ctrl+C to stop
 1   192.168.5.1   15.262 ms  15.773 ms  15.394 ms
 2   10.1.35.1   76.135 ms  61.636 ms  75.652 ms
 3   10.1.13.1   76.793 ms  77.909 ms  77.061 ms
 4   10.1.12.2   75.936 ms  75.886 ms  75.085 ms
 5     **192.168.2.3   90.732 ms (ICMP type:3, code:3, Destination port unreacha
ble)

PC5> trace 192.168.6.2
trace to 192.168.6.2, 8 hops max, press Ctrl+C to stop
 1   192.168.5.1   14.839 ms  15.907 ms  15.141 ms
 2   10.1.35.1   75.488 ms  75.762 ms  75.571 ms
 3   10.1.13.1   76.281 ms  74.499 ms  75.270 ms
 4   10.1.12.2   75.392 ms  76.381 ms  77.268 ms
 5   10.1.24.2   76.181 ms  76.046 ms  90.915 ms
 6   10.1.46.2   106.495 ms  105.244 ms  106.041 ms
 7     *  *  *
 8   *192.168.6.2   136.016 ms (ICMP type:3, code:3, Destination port unreachabl
e)

PC5>
```

Ping Test Results:

PC6 (192.168.6.2)

```
PC6> ping 192.168.1.3
192.168.1.3 icmp_seq=1 timeout
84 bytes from 192.168.1.3 icmp_seq=2 ttl=60 time=91.113 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=60 time=122.114 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=60 time=105.445 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=60 time=123.585 ms

PC6> ping 192.168.2.3
192.168.2.3 icmp_seq=1 timeout
192.168.2.3 icmp_seq=2 timeout
84 bytes from 192.168.2.3 icmp_seq=3 ttl=61 time=90.940 ms
84 bytes from 192.168.2.3 icmp_seq=4 ttl=61 time=91.667 ms
84 bytes from 192.168.2.3 icmp_seq=5 ttl=61 time=90.711 ms
```

- **Ping to PC2 (192.168.1.3):** First packet timed out (icmp_seq=1), then succeeded from icmp_seq=2 onward with TTL=60 and latency ~91–123 ms. The initial timeout is expected — it's the ARP resolution delay on first contact.
- **Ping to PC4 (192.168.2.3... likely a typo, actually reaching R2/PC3's subnet):** First two packets timed out, then succeeded from icmp_seq=3 with TTL=61 and latency ~90–92 ms.

PC1 (192.168.1.2)

```
PC1> ping 192.168.2.1
84 bytes from 192.168.2.1 icmp_seq=1 ttl=254 time=44.786 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=254 time=44.948 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=254 time=45.856 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=254 time=46.963 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=254 time=45.077 ms

PC1> ping 192.168.6.2
192.168.6.2 icmp_seq=1 timeout
192.168.6.2 icmp_seq=2 timeout
84 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=122.276 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=122.910 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=120.697 ms

PC1>
```

- **Ping to R2 (192.168.2.1):** All 5 packets succeeded immediately with TTL=254 (indicating it's hitting the router interface directly — routers typically use higher default TTLs) and low, stable latency (~44–47 ms).
- **Ping to PC6 (192.168.6.2):** First two packets timed out, then succeeded from icmp_seq=3 with TTL=60 and latency ~120–122 ms.

Challenges faced:

1. OSPF Multi-Area Design:

It was important to plan out R1, R2 and R4 so that they would be functioning as Area Border Routers (ABRs) correctly, with consistent adjacencies across their respective areas, without creating routing loops or area misconfigurations when setting up OSPF for three different areas (Area 0, Area 1 and Area 2).

2. Distribution of the Protocol:

Since the network used OSPF, RIPv2, and static routing, redistribution between protocols (particularly at R3, where OSPF and RIPv2 meet) required manual metric assignment to avoid routing inconsistencies and to ensure that routes were properly advertised in both directions without unintended route flapping.

3. Floating Static Backup Pathways

The two backup serial links (R1–R4 and R5–R4) were configured as floating static routes with the correct administrative distances (150), so they would only be used if the primary OSPF/RIP paths failed. These backups had to be tested repeatedly to confirm that they only kicked in during simulated link failures and not during normal operation.

4. GNS3 Hardware Specific Problems

One of the first hurdles was that an ESW1 device that was supposed to be a VLAN capable switch was running a pure router IOS with no Layer 2 switching capabilities. This was changed to a proper Ethernet switch node and full VLAN functionality restored.

5. Uniform IP Addressing and Documentation

As the topology became more complex, it was necessary to maintain a disciplined documentation of an accurate and up-to-date addressing scheme across six routers, six point-to-point links, two backup links, and four LANs to avoid subnetting conflicts or duplicate IP assignments.

Conclusion:

The project successfully designed, implemented and verified a multi-protocol routing environment that incorporates OSPF, RIPv2 and static routing within a single simulated network using GNS3. The topology was designed to simulate a real-world enterprise environment where multiple routing protocols are used and need to be redistributed to communicate with each other. The network is divided into three OSPF areas and RIPv2 is used in addition to static and floating static routes.

Testing confirmed the ability of all four LANs to communicate end-to-end, with routing tables, OSPF neighbour relationships, and redistribution behaviour verified via direct router output. Results from ping and traceroute indicated that traffic was taking the correct routes with hop counts and latency values consistent with the physical and logical distance between networks. This project overall reiterated important networking concepts such as OSPF multi-area design, route redistribution between dynamic and static routing protocols, floating static routes for redundancy, and practical troubleshooting skills to diagnose and fix simulation specific issues.